

GlobalGeoTree: Source Bias and Temporal Development

RQ2 - Source and Platform Bias, and RQ3 - Temporal Development, combined into one report

Research Questions

This report combines two related research questions, because they share the same data and the same key idea: **GlobalGeoTree is not a neutral, evenly-collected dataset** - it is built from thousands of different sources (mostly citizen-science apps), and that has consequences for how the geographic and temporal patterns in it should be read.

RQ2 - Source and Platform Bias. To what extent do data sources shape the observed geographic and taxonomic patterns in GlobalGeoTree?

- 1. Which sources contribute the largest share of observations?
- 2. Is the dataset dominated by citizen-science platforms such as iNaturalist?
- 3. Do different sources cover different countries or regions?
- 4. Are some sources associated with specific years, countries, or species groups?

RQ3 - Temporal Development. How has the number and distribution of tree observations changed from 2015 to 2024?

- 1. Has the number of observations increased over time?
- 2. Which years show the strongest growth or decline?
- 3. Did geographic coverage expand over time?
- 4. Are temporal patterns mainly driven by specific sources?

Both analyses are descriptive. Strong source-specific patterns should be read as evidence of recording and platform bias, not only as biological patterns in tree distributions.

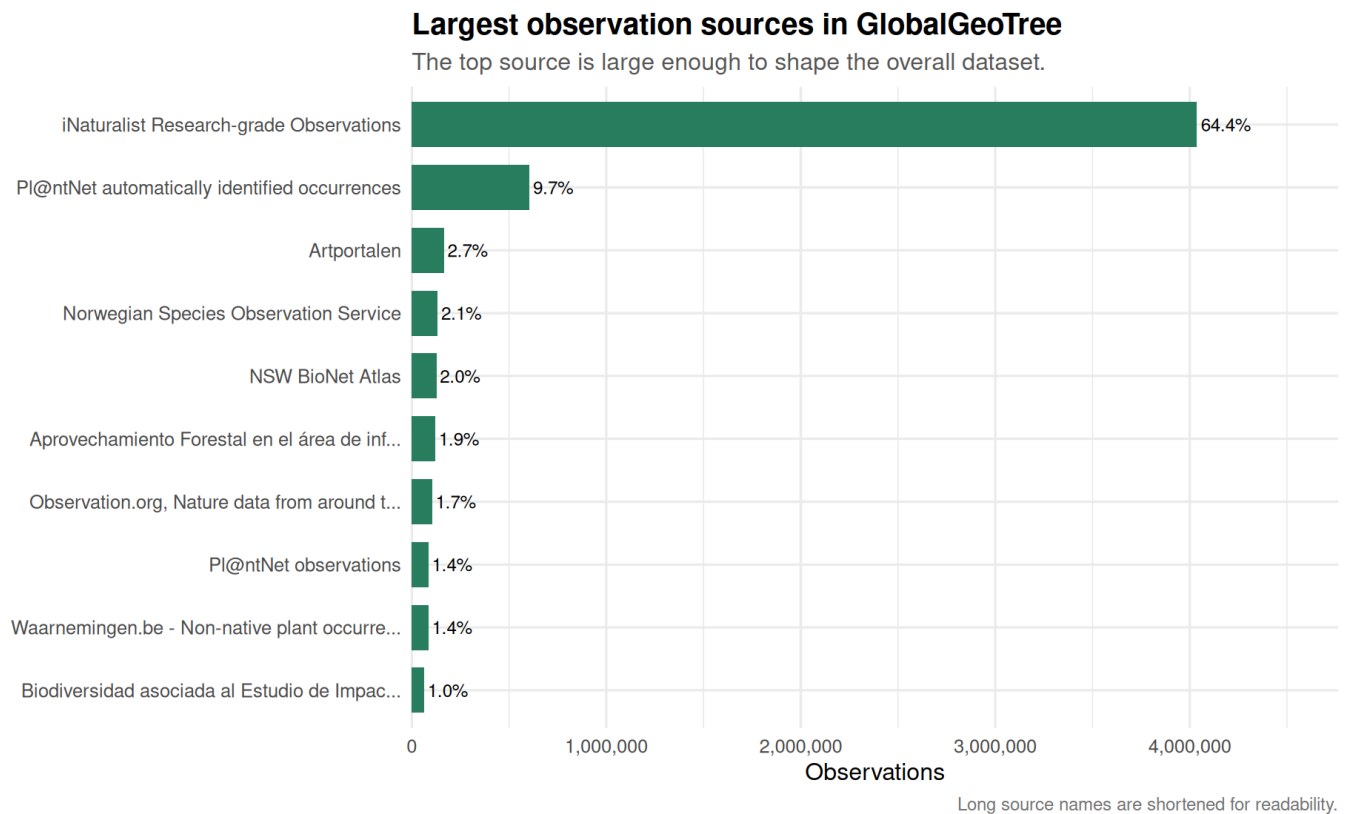
Data Loading

Measure	Value
Observation file	GlobalGeoTree.csv
Period analysed	2015-2024
Rows analysed	6,263,345
Distinct sources	2,493
Distinct source groups	5
Distinct countries	221
Distinct species	21,001

This report uses 6,263,345 observations.

Source Concentration

Why this visualisation is needed. Before comparing countries, years, or species, we first need to know whether the dataset is balanced across sources. If one source is extremely large, it can shape the whole dataset.

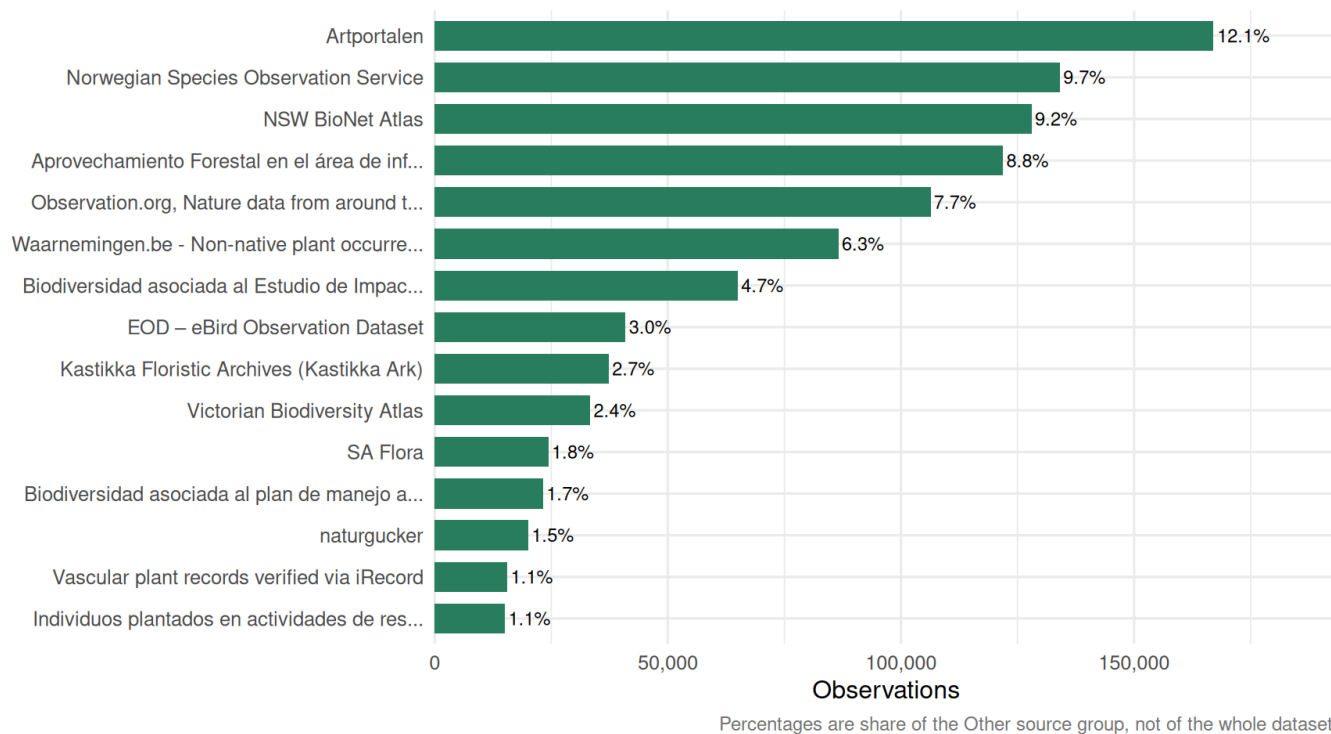


What Is in “Other Source”?

Why we look at this: “Other source” is the second-largest group in this dataset, but unlike iNaturalist or Pl@ntNet it is not one platform - it is everything that didn’t match a recognised keyword pattern. Before using it in any chart, it’s worth showing what’s actually inside it.

The 15 largest individual sources inside "Other source"

"Other source" is made up of 2,330 differently-named sources in total.



Interpretation. “Other source” is not a single platform - it is a long tail of 2,330 individually named sources (national biodiversity portals, regional/government forestry or environmental-impact datasets, local citizen-science platforms such as Artportalen or Observation.org, and similar projects). Treat any chart that shows “Other source” as a group as a *mixture*, not as one consistent collection method - the countries and years where “Other source” is large are typically being driven by a handful of large national datasets rather than a single shared bias.

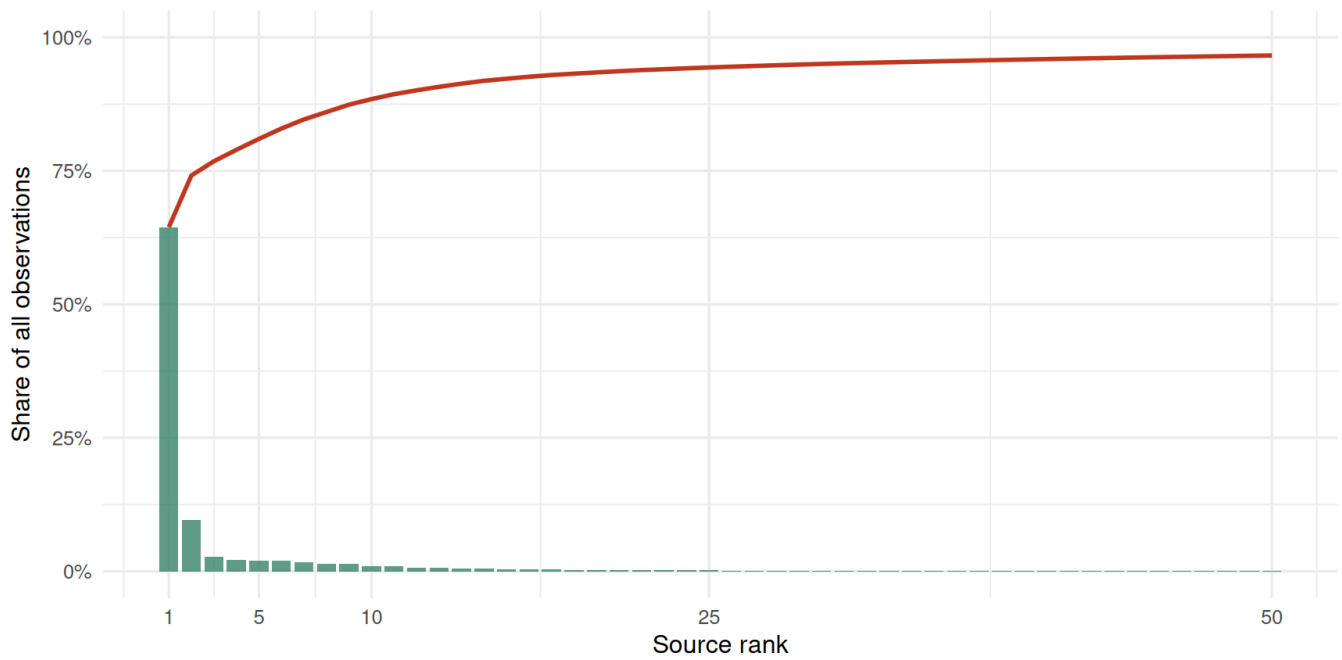
Source Concentration

Why we look at this: before comparing countries, years, or species, we first need to know whether the dataset is balanced across sources. If one source is extremely large, it shapes the whole dataset (RQ2 subquestions 1-2).

The source structure is highly concentrated: **iNaturalist alone contributes 64.5%** of observations, and **iNaturalist plus Pl@ntNet together contribute 75.5%**.

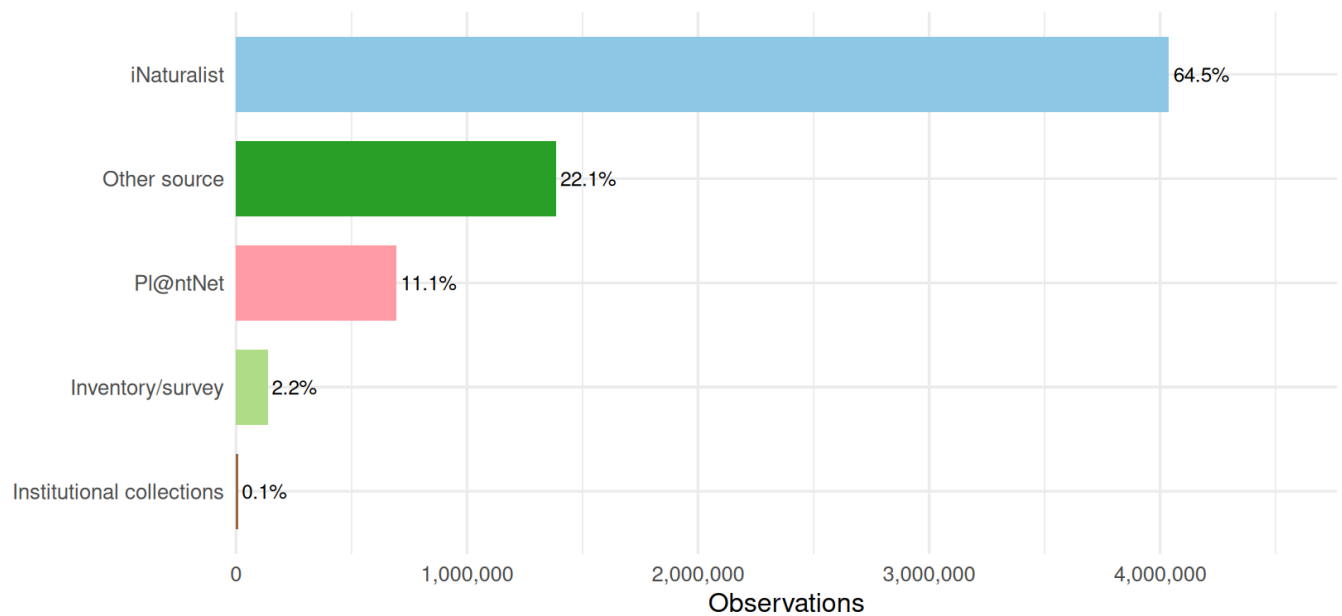
A small number of sources explain most observations

Bars show each source's individual share; the red line shows cumulative share.



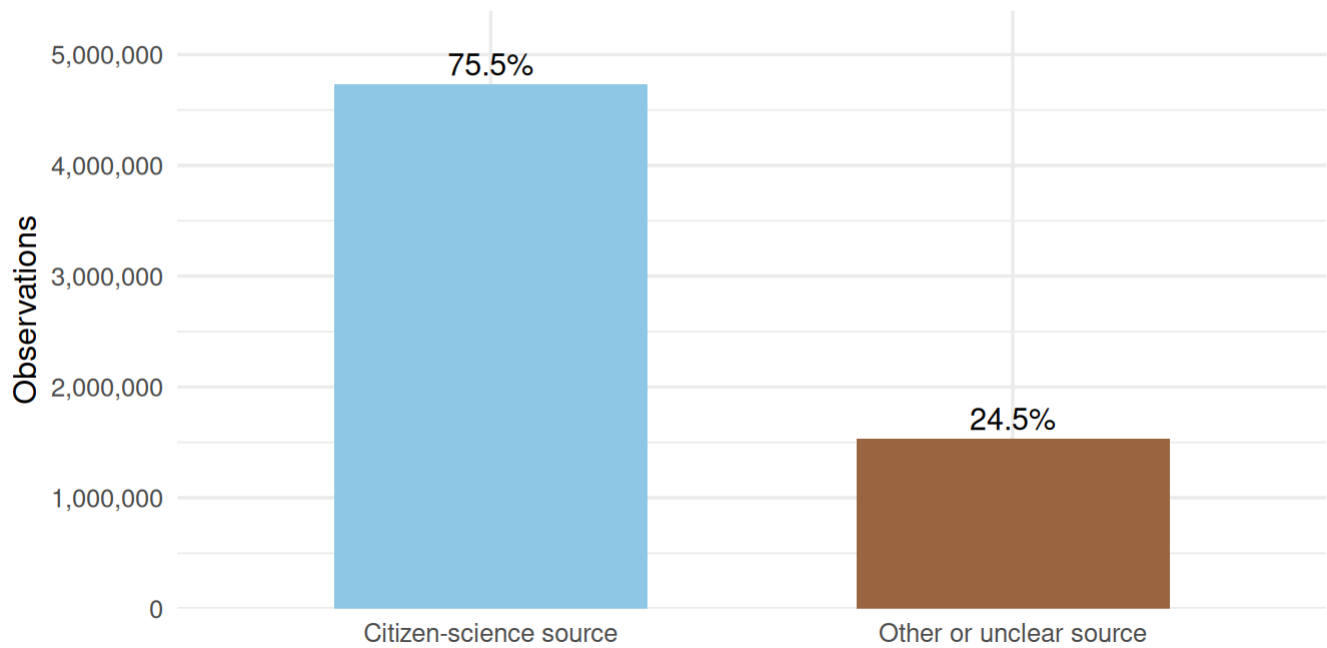
Observation share by broad source group

Citizen-science platforms dominate the grouped source structure.



Citizen-science platforms dominate the observations

Citizen-science is defined here as iNaturalist plus Pl@ntNet.



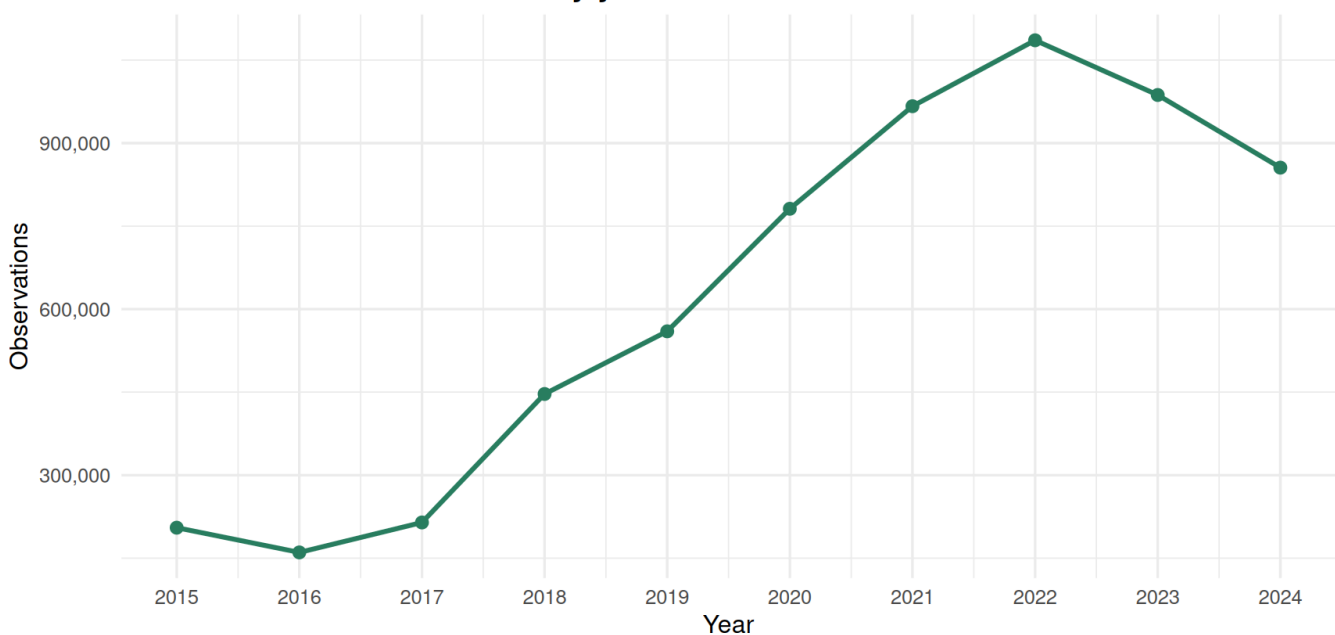
Interpretation. A few large platforms dominate the evidence base, so global summaries are strongly affected by the behaviour, geography, and taxonomy covered by those platforms specifically.

Temporal Development

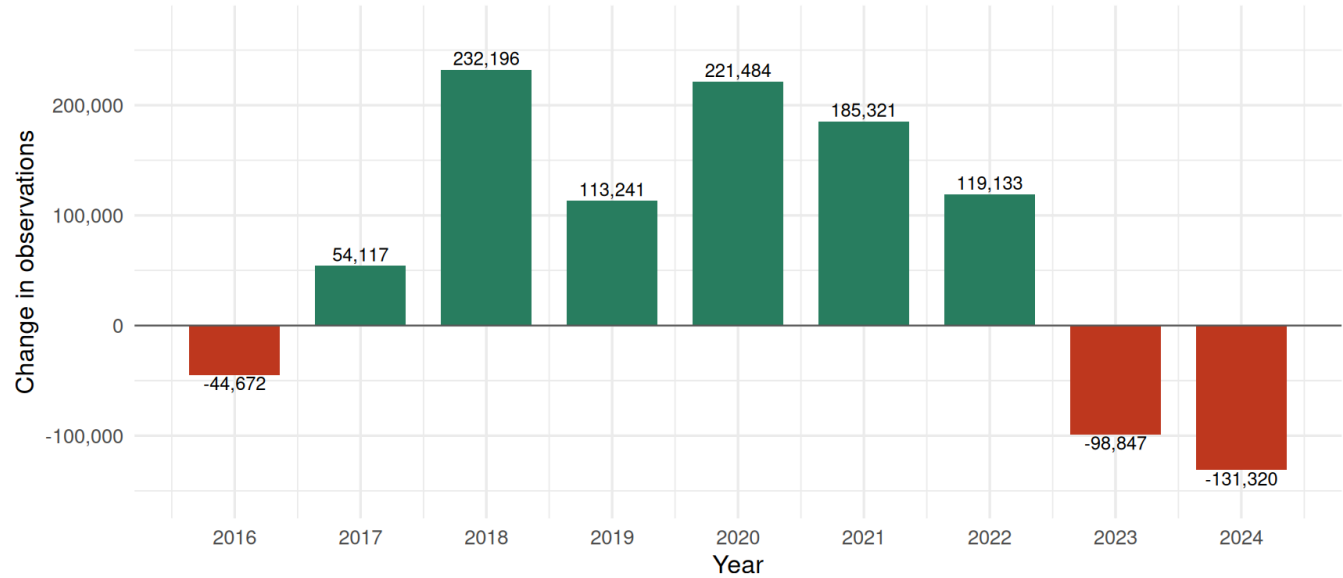
Why we look at this: before anything more detailed, we need the basic shape of the data over time - is it going up, down, or flat (RQ3 subquestions 1-2) - and whether that shape is really about trees, or about one platform growing (RQ2 subquestion 4 / RQ3 subquestion 4).

Between 2015 and 2024, observations changed from **205,055** to **855,708**, an overall change of **317.3%**.

Number of tree observations by year



Year-to-year change in observations

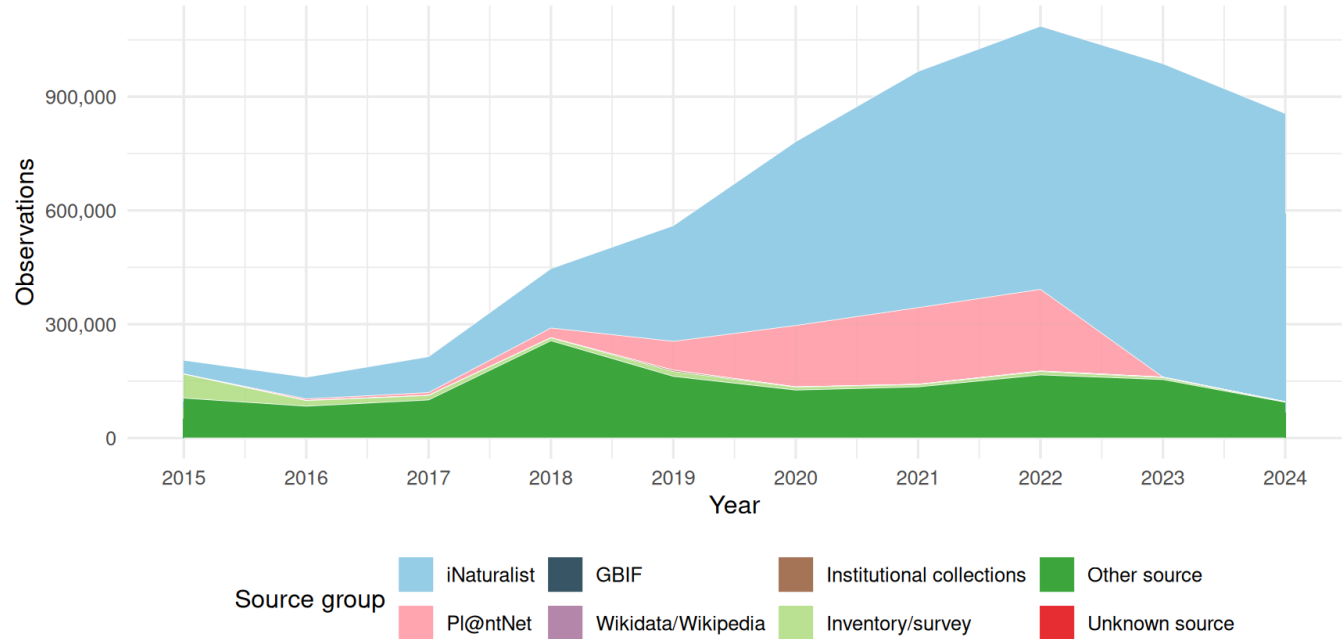


Note: 2024 is the last year in the export and may not be complete yet, so part of the apparent decline could be a data-cutoff effect rather than a real drop in recording.

Interpretation. The strongest growth occurs in **2018** (+232,196), and the strongest decline in **2024** (-131,320). The 2024 figure should be treated with caution: if that year’s data was exported before the year finished, the apparent decline may shrink once a complete year is available.

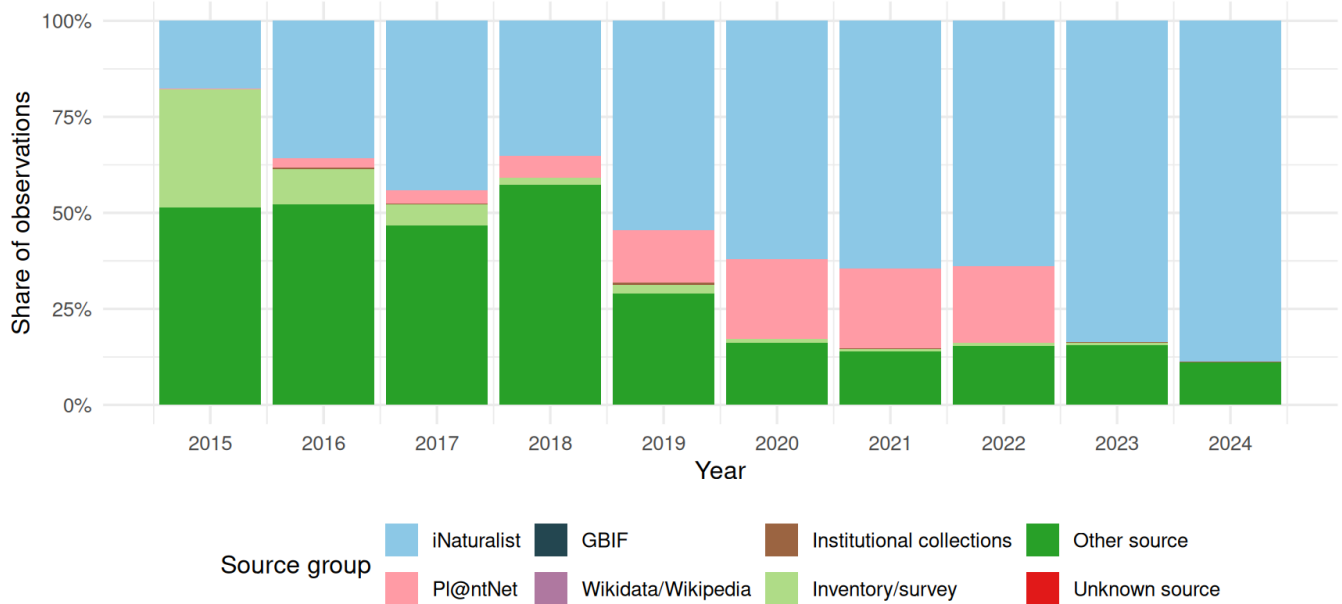
Observation volume by source group and year

A rising source group can drive the apparent growth of the full dataset.



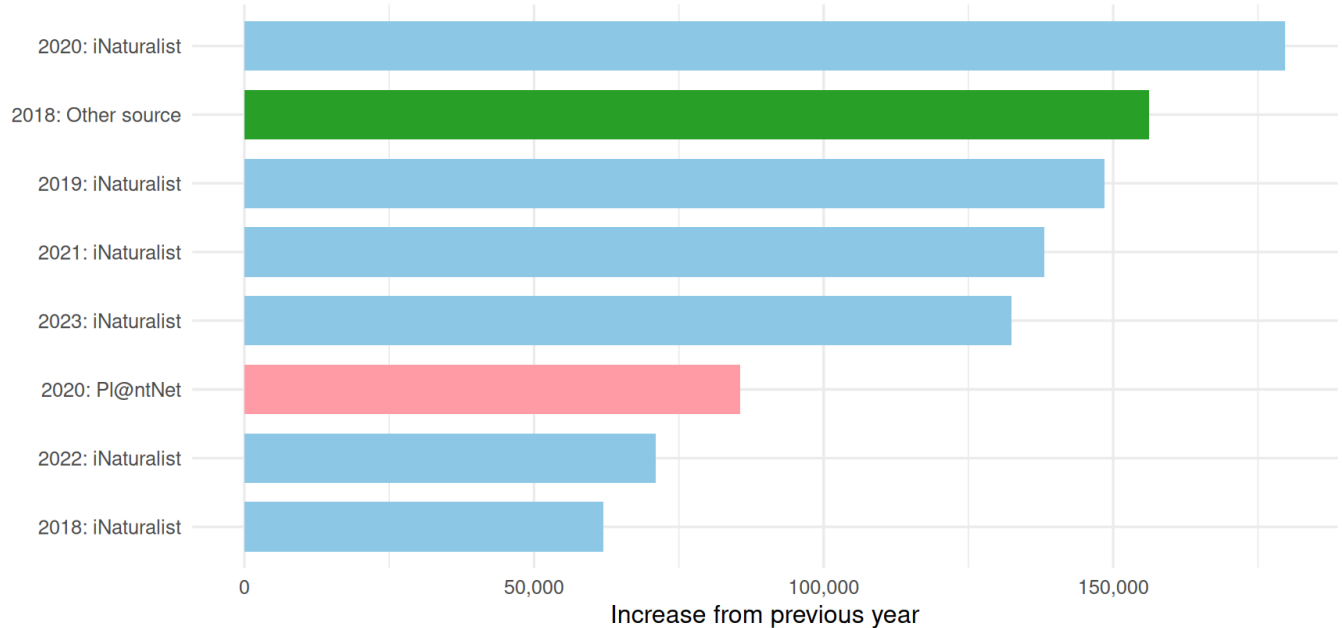
Source composition by year

Bars sum to 100% within each year.



The most recent year may be incomplete depending on the data export date.

Largest source-specific yearly increases

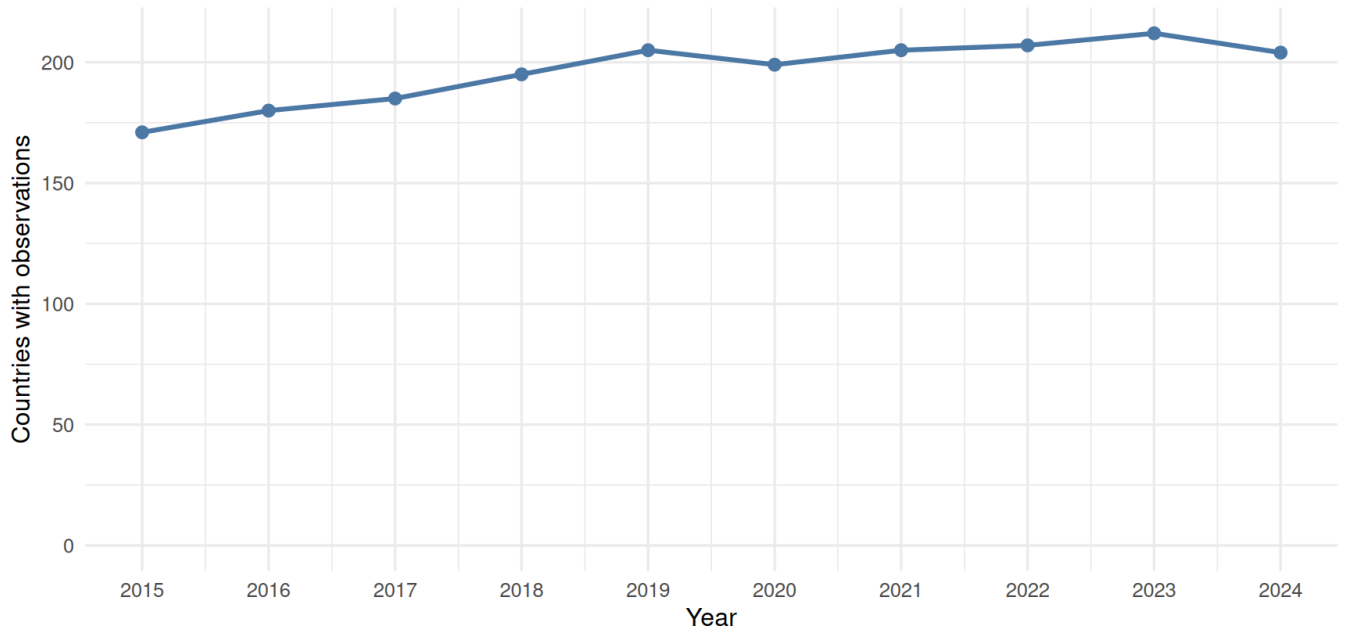


Interpretation. If most growth is concentrated in one source group, the temporal trend is source-driven, not a neutral increase in global tree recording. This ranks *absolute* increases, so later years (when the whole dataset is bigger) have a built-in advantage - a percentage-growth version would be a useful follow-up.

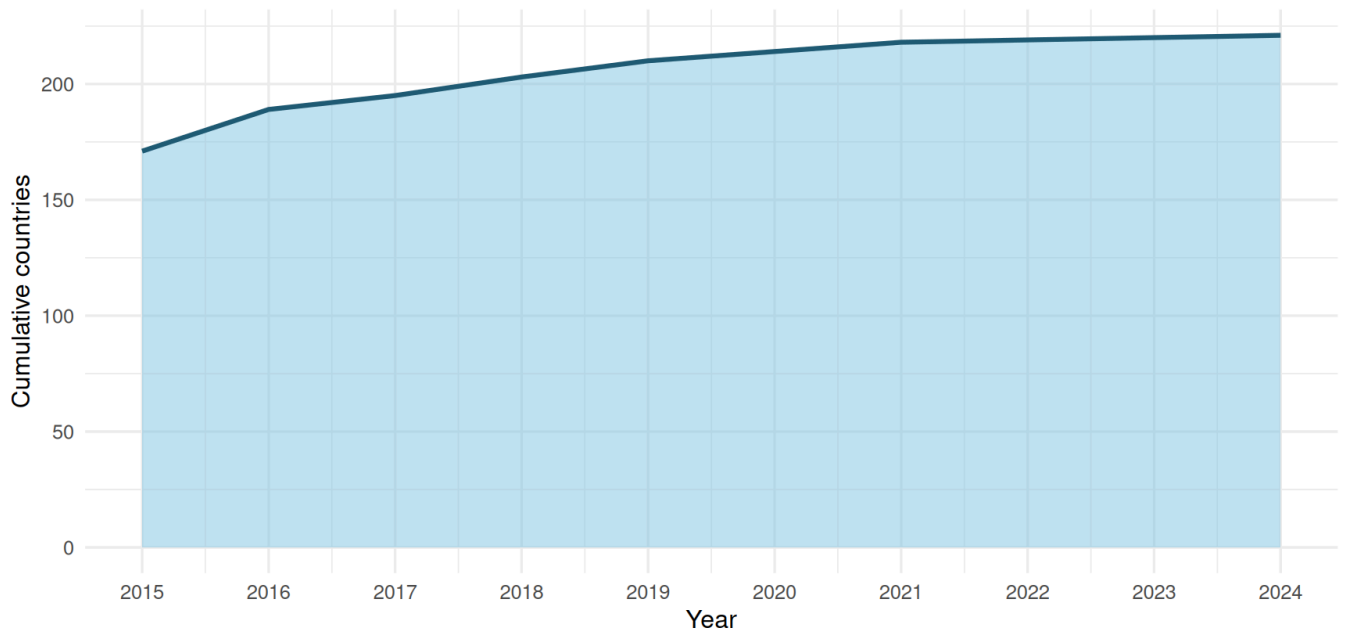
Geographic Coverage Over Time

Why we look at this: more observations is not the same as broader observations - growth could just mean recording the same countries more intensively (RQ3 subquestion 3).

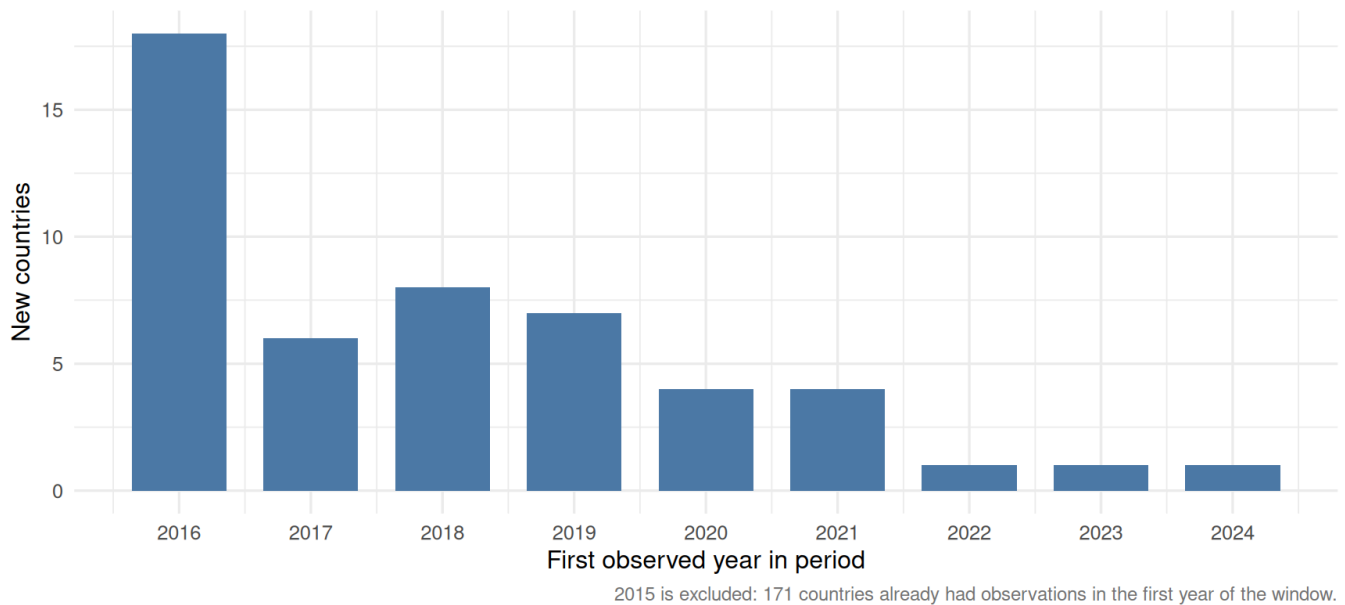
Countries represented each year



Cumulative country coverage over time



New countries entering the dataset by year



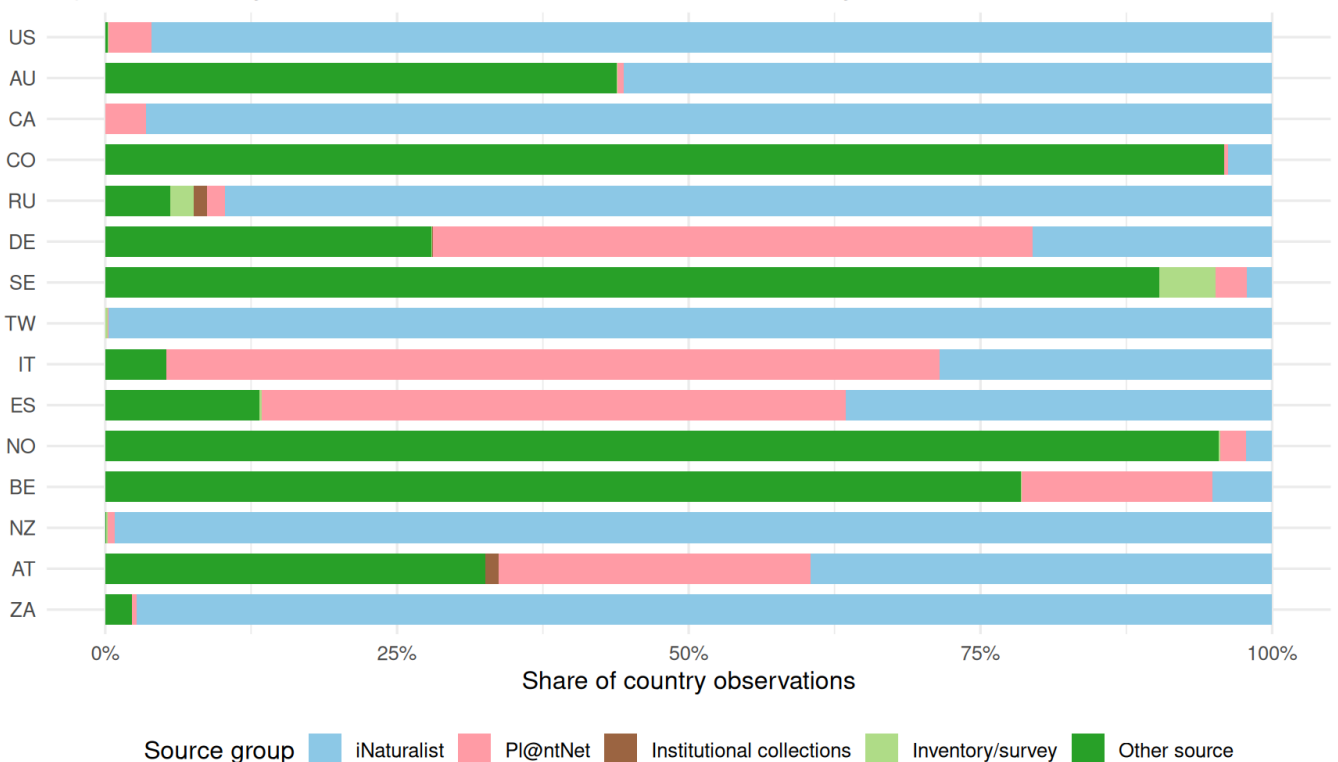
Interpretation. If country counts increase over time, GlobalGeoTree is expanding geographically; if observations grow without much country expansion, growth is concentrated in already well-covered places.

Countries and Regional Coverage

Why we look at this: a global source pattern can hide strong differences between countries - is each country's evidence base built the same way (RQ2 subquestion 3)?

Source composition differs by country

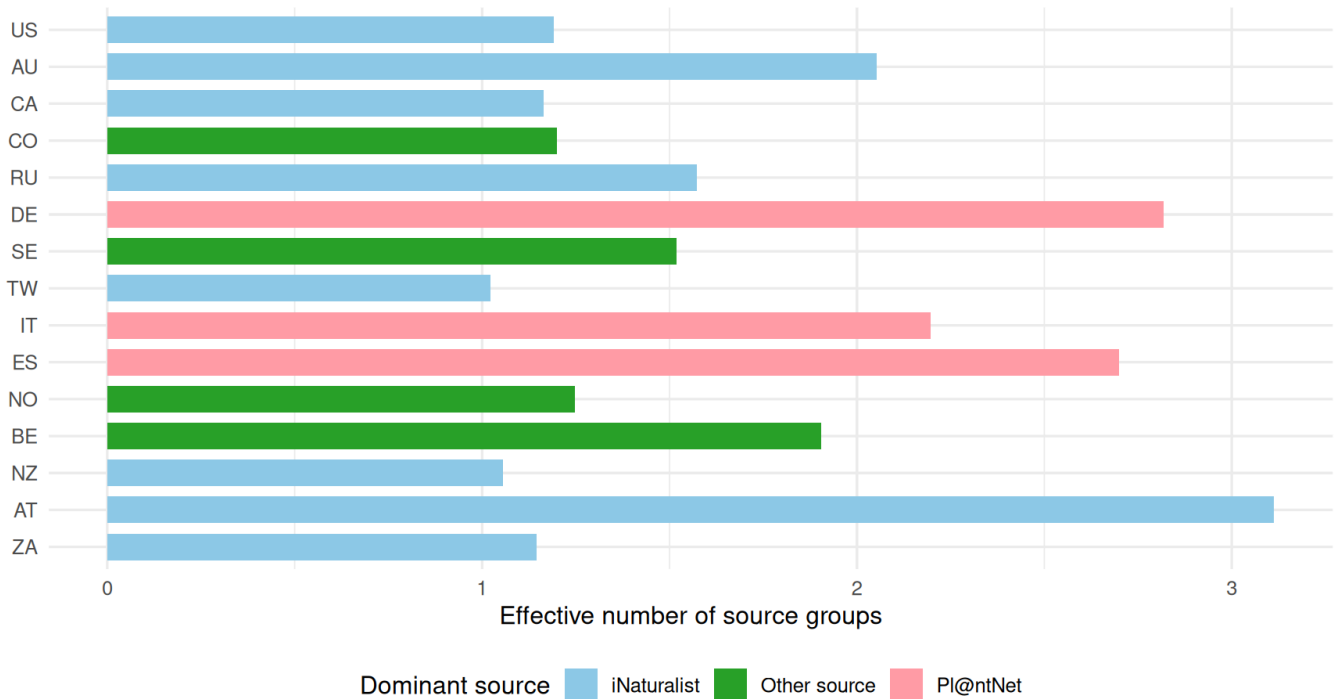
Top 15 countries by observations; bars sum to 100% within each country.



Interpretation. Countries differ both in volume and in source mix - a country dominated by iNaturalist is not directly comparable to one dominated by local inventories or unknown sources.

How source-diverse are the largest country samples?

Effective number of source groups is low when one source dominates a country.



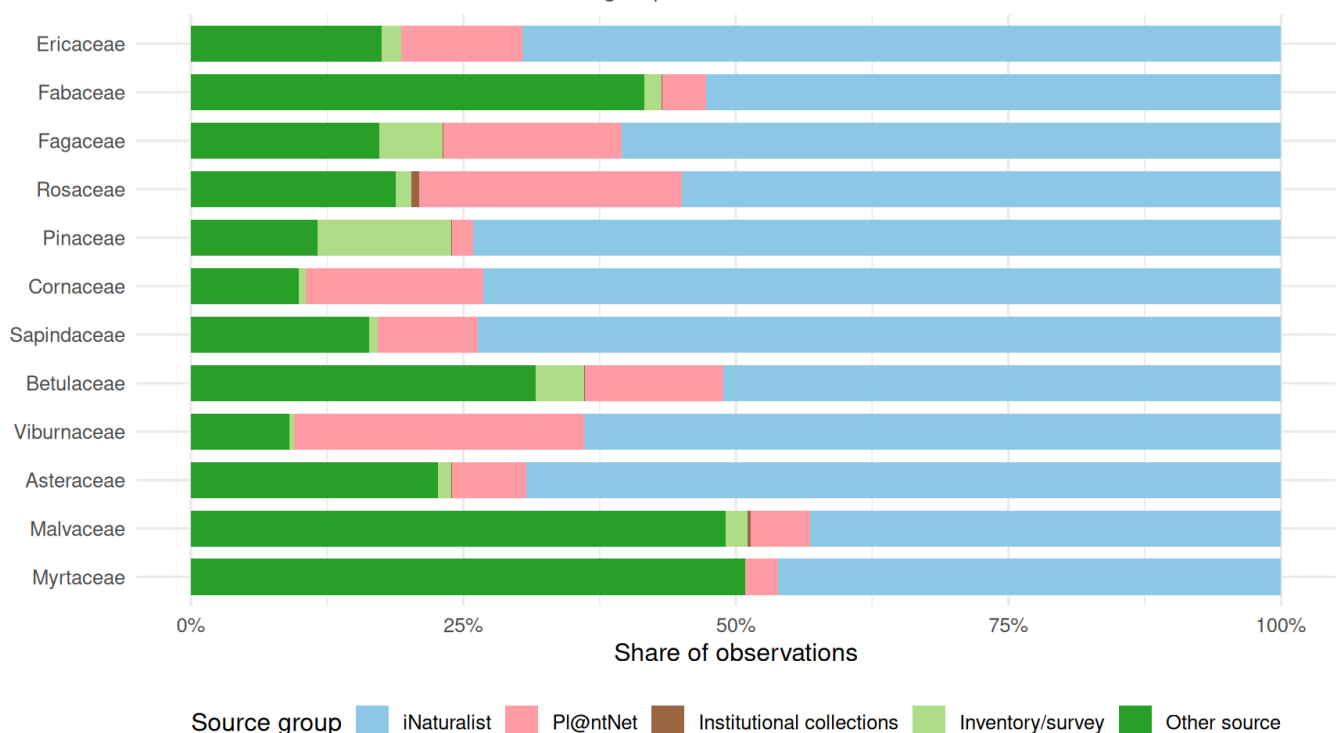
Interpretation. Low diversity means a country's evidence base is essentially one platform's user base, so any pattern found there is at higher risk of being platform-specific rather than biological.

Sources and Taxonomic Groups

Why we look at this: source bias can also be taxonomic - some platforms may record ornamental or easy-to-photograph families more often than others (RQ2 subquestion 4).

Source composition by family

Bars sum to 100% within each taxonomic group.



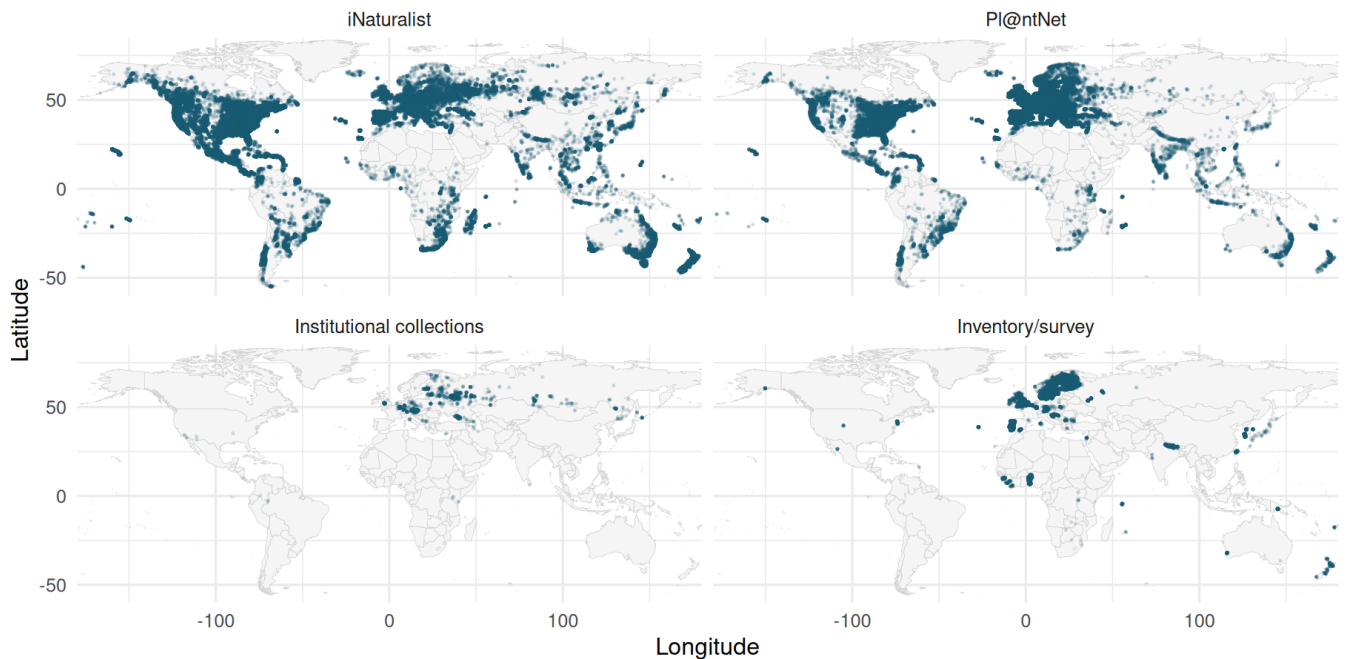
Interpretation. If a family is mostly recorded by one platform, its geographic pattern may partly reflect that platform's user base rather than the tree's real distribution.

Geographic Footprints of Sources and Time

Why we look at this: spatial clusters in the raw points may reflect user activity and platform coverage, not only real tree distributions (RQ2 subquestion 3 and RQ3 subquestion 3, viewed spatially).

Geographic coverage of major source groups

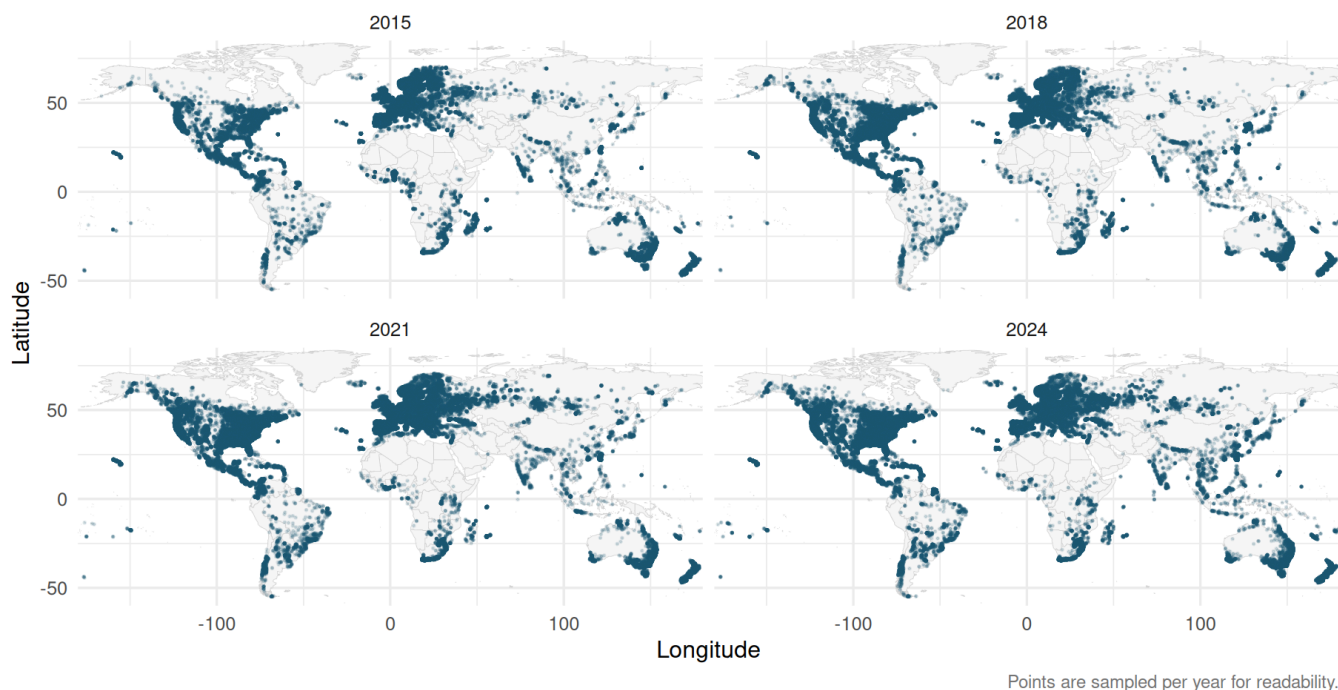
Different platforms leave visibly different spatial footprints.



Points show recording locations, not true species ranges. Points are sampled for readability.

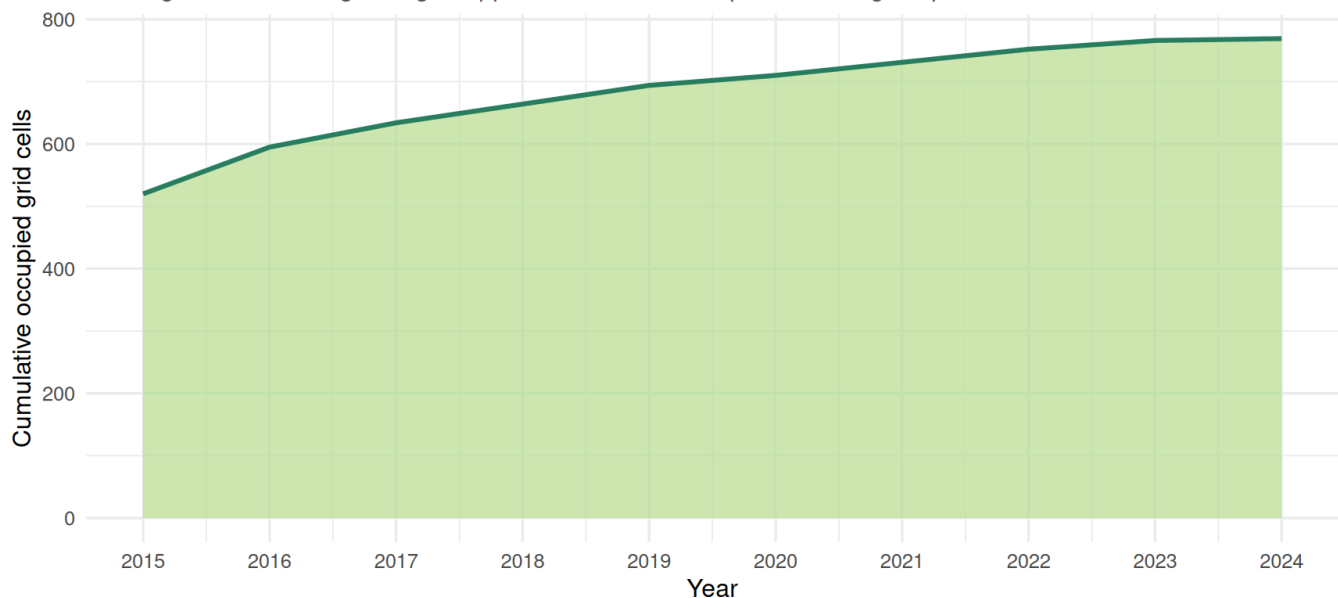
Map snapshots of observations over time

Selected years show whether records spread geographically or intensify in existing regions.



Cumulative spatial grid coverage over time

A 5-degree latitude/longitude grid approximates whether spatial coverage expands.



Interpretation. Source maps and map snapshots distinguish two different processes: where each platform records data, and whether records over time spread into new regions or just intensify in existing ones.

Unknown Source Diagnostics

Out of 6,263,345 observations, only 0 (0.00%) come from an unknown or missing source. This is too small a group to break down by country - **metadata quality is not a meaningful concern for this dataset**, so source-specific patterns elsewhere in this report are not undermined by missing source labels.

Overall Interpretation

This analysis shows that GlobalGeoTree is **not source-neutral and not time-neutral**:

- A few platforms - mainly iNaturalist and Pl@ntNet - dominate the observation base.
- “Other source” looks big in some charts, but it is a long tail of many different national/regional datasets, not one consistent collection method (see the dedicated section above).
- Source composition varies by country, year, and taxonomic family, so geographic and taxonomic patterns are partly platform-coverage patterns.
- The overall rise in observations from 2015-2024 is driven substantially by iNaturalist’s growth, and the apparent decline in 2024 may be a data-export artifact rather than a real drop.
- Geographic coverage genuinely expanded (more countries, more grid cells over time), but that expansion is also unevenly distributed across platforms.

Main limitation. These figures describe patterns in the available records. They do not prove that any source is biased in a negative way - they show where source coverage is uneven and where interpretation needs extra care. For any downstream analysis, source composition should be reported alongside geographic, temporal, or taxonomic results, and key findings should ideally be re-checked with and without the largest citizen-science source.